High Finance — Full Feature Audit

What It Is

A full-stack investment research platform for investors and finance students. Aggregates public company data (fundamentals, prices, 13F ownership, insider/congressional trades, earnings, options) and layers on quantitative tools (DCF, Monte Carlo, Black-Scholes, ML) plus AI research assistance (memos, earnings analysis, conversational Q&A).

Stack: FastAPI + PostgreSQL + Redis | React/TypeScript | Railway + Vercel

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Where It Exceeds Competitors

Strength	Who It Beats	Why
Congressional trade backtesting with win rates	Bloomberg, Quiver Quant, Unusual Whales	Nobody else ranks politicians by backtested return. Quiver shows raw data; you show performance.
Interactive DCF with market-implied divergence detection	Finviz, Koyfin, most paid tools	Detecting when user assumptions diverge from market-implied values and letting you toggle is rare even in paid tools.
Monte Carlo risk simulation in a browser	Yahoo Finance, Finviz, Seeking Alpha	VaR, max drawdown, earnings shock stress tests are typically locked behind $24k/yr terminals.
Investment project workspaces with DCF version history	Every free tool	Most give you a watchlist. You give a research workflow: thesis → catalysts → risks → peers → DCF snapshots. That's how buy-side analysts actually work.
AI earnings analysis (8-K press release parsing)	Yahoo Finance, Finviz, TradingView	Auto-extracting guidance, sentiment, key metrics from filings. Usually requires Sentieo/AlphaSense ($10k+/yr).
Options decay simulator with per-strike Greeks	Robinhood, Webull, Yahoo Finance	Most free tools show option chains but don't let you simulate theta decay over time with a visual chart.
Ticker bundle endpoint (single call, ~20 data streams)	N/A (architectural)	Smart design: parallel fetch with graceful degradation. Reduces perceived latency significantly.

Weaknesses — With Specific Improvements

1. Data Foundation (Critical)

Problem: yfinance is a scraper, not a data provider. No SLA, rate limits, breaks without notice.

Improvements:

- **Short-term:** Add data freshness timestamps to every card in the UI ("Last updated: 2 min ago" vs "Last updated: 3 days ago"). Users need to know when data is stale.
- **Short-term:** Add health monitoring — if yfinance returns errors or stale data, surface it in the admin dashboard and optionally to users ("Price data temporarily unavailable").
- **Medium-term:** Abstract data fetching behind a provider interface so you can swap yfinance for a licensed feed (Polygon.io at $29/mo, Twelve Data, Alpha Vantage) without rewriting services.

- **Long-term:** License a real data feed. Polygon.io's starter plan gives real-time US equities for $29/mo.

2. ML Predictions (Liability)

Problem: Basic features (RSI, volume ratio) predicting next-day direction is effectively random. Showing this to finance students teaches bad habits.

Improvements:

- **Option A:** Remove the feature entirely. Honest and low-effort.
- **Option B:** Reframe as "educational" — show feature importance, explain why naive ML fails on markets, use it as a teaching tool about overfitting and market efficiency. Add heavy disclaimers.
- **Option C:** Replace with factor-based analysis (value, momentum, quality scores) which has actual academic backing. Show factor exposures rather than point predictions.

3. No Real-Time Data

Problem: No WebSocket streaming. Prices and options are stale snapshots.

Improvements:

- **Short-term:** Show "as of" timestamps prominently. Add a "Refresh" button per card.
- **Short-term:** Auto-refresh key data (price, options) every 60s during market hours using the market hours detection pattern you already have.
- **Medium-term:** Add WebSocket price streaming for watchlist tickers (Polygon.io WebSocket or Finnhub free tier).
- **Long-term:** Real-time options flow would be a killer feature but requires expensive data ($$$).

4. No Collaboration

Problem: Projects are per-user. Can't share research with classmates or mentors.

Improvements:

- **Short-term:** "Export to PDF" for project workspaces. Shareable without accounts.
- **Medium-term:** Read-only sharing via link (generate a UUID, serve a public read-only view).
- **Long-term:** Multi-user projects with roles (owner, editor, viewer). Comments on DCF assumptions.

5. No Strategy Backtesting

Problem: You backtest politician trades but not user-defined strategies or screens.

Improvements:

- **Short-term:** Add historical performance to saved screener filters — "Stocks matching this screen 1 year ago returned X% on average."
- **Medium-term:** Simple backtest engine: "Buy when screen matches, sell after N days, show aggregate stats."
- **Long-term:** Custom strategy builder with rule-based entries/exits.

6. Shallow Technical Analysis

Problem: Chart supports drawings but no indicator overlays.

Improvements:

- **Short-term:** Add SMA/EMA overlays (20, 50, 200-day) — the most requested indicators. These are trivial to compute from existing price data.
- **Medium-term:** Volume bars below price chart. MACD and RSI as sub-charts.
- **Note:** Don't try to compete with TradingView's 100+ indicators. Pick the 5-10 most used and do them well.

7. AI Features Feel Bolt-On

Problem: AI assistant waits for questions rather than surfacing insights proactively.

Improvements:

- **Short-term:** Add contextual AI prompts — when viewing a DCF, show a suggestion like "Ask AI: Is this growth rate realistic given the industry?" as clickable chips.
- **Medium-term:** Auto-generate a 2-sentence insight per ticker on the overview tab ("Revenue growth is decelerating while margins expand — typical of a maturing SaaS company").
- **Medium-term:** AI-powered screener — natural language query ("Show me profitable small-caps with insider buying") translated to screener filters.
- **Long-term:** Proactive alerts ("Your watchlist stock AAPL just reported earnings that missed estimates — here's what changed").

8. Underdeveloped Alerts/Notifications

Problem: Only congress trade notifications exist. No price, earnings, or insider alerts.

Improvements:

- **Short-term:** Earnings date reminders for watchlist stocks (you already have earnings calendar data).
- **Medium-term:** Price threshold alerts (52-week high/low, % change) — store in DB, check in a scheduled RQ job.
- **Medium-term:** Insider buying alerts for watchlist stocks.
- **Long-term:** Email/push notification delivery.

Missing Features — Prioritized Roadmap

Tier 1: High Impact, Builds on Existing Data

Feature	What It Adds	Existing Foundation
Historical financial charting	Revenue, margins, EPS as interactive line charts over 10 years	You already fetch 10yr financials — just need visualization
Peer comparison visualization	Side-by-side multi-company charts and metrics	Comps data exists in valuation service
Portfolio-level analytics	Sharpe ratio, beta, sector concentration, correlation matrix	Portfolio positions already tracked
Data freshness indicators	"Last updated X ago" on every card	Timestamps exist in cache layer
Auto-refresh during market hours	Price/options refresh every 60s when market is open	Market hours detection already implemented

Tier 2: Medium Impact, Moderate Effort

Feature	What It Adds	Notes
Dividend analysis	Yield history, payout ratio trends, ex-date tracking, dividend growth rate	yfinance provides dividend history
Short interest data	Days to cover, short % of float, short squeeze risk score	Requires new data source (Finviz scrape or API)

Analyst ratings view	Consensus rating, price target range, rating history	Partially exists in DCF analyst overlay
Compare mode	`/compare/AAPL/MSFT` — side-by-side across all metrics	New route + reuse existing cards
Export to PDF	Research project → professional report	React-to-PDF libraries exist
Natural language screener	"Profitable small-caps with insider buying" → filters	Claude API + existing screener filters
Economic calendar	FOMC, CPI, jobs report dates with market impact notes	New data source needed

Tier 3: High Impact, High Effort (Future Moat)

Feature	What It Adds	Notes
Strategy backtesting	"If I bought when this screen matched, what would returns be?"	Needs historical screener snapshots or point-in-time data
Real-time price streaming	WebSocket updates for watchlist tickers	Polygon.io or Finnhub WebSocket
Collaborative projects	Share research, invite reviewers, comment on assumptions	Auth + permissions + sharing infrastructure
ETF exposure analysis	"Which ETFs hold AAPL?" + sector/factor exposure breakdown	New data source (ETF holdings databases)
Mobile app	Native iOS/Android	Could use React Native or Capacitor to reuse web code
Community features	Follow analysts, share research, prediction leaderboard	Major product expansion

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Competitive Positioning

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                    Free ←──────────────────────→ Paid ($200+/mo)

  Yahoo      Finviz      HIGH FINANCE      Koyfin       Bloomberg
  Finance                                               Terminal
  ────────   ──────────  ────────────────  ──────────   ──────────
  Broad      Screening   Deep analysis     Charting +   Everything
  Shallow    + heatmaps  + AI + Congress   Fundamentals
  Ads        Static      Interactive       Multi-company
                         Project-based     Time-series
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Your unique wedge: AI-integrated equity research workflow with political trading intelligence. Nobody else combines DCF modeling, Monte Carlo risk, AI earnings analysis, and congressional trade backtesting in one free platform.

Biggest risk: Data reliability (yfinance dependency).

Strategic advice: Don't chase breadth (TradingView's indicators, Bloomberg's data). Chase depth in your unique areas:

1. Make the AI research workflow best-in-class (proactive insights, natural language queries)

2. Expand political/institutional trading intelligence (committee assignments → trade correlation, timing analysis)
3. Add collaboration for the student use case (shared projects, exportable reports)
4. License a real data feed before scaling users

Quick Wins (< 1 day each)

1. Add "Last updated" timestamps to all data cards
2. Auto-refresh prices every 60s during market hours
3. Add SMA overlays (20/50/200) to price chart
4. Show earnings date countdown on watchlist
5. Add clickable AI prompt suggestions on ticker pages
6. Historical revenue/earnings as line charts (data already fetched)